

A three-resource starter kit for TLA+

In distributed architectures, subtle edge cases around concurrency and consensus are hard to surface through conventional testing alone. **Formal specification** gives teams a clear way to validate system designs, align around verified invariants, and de-risk core infrastructure before deployment. These three accessible resources offer a solid foundation.

1 Foundations · Leslie Lamport's video course

Lamport introduces state machines, temporal logic, and system behavior in an approachable, step-by-step manner.

Overview lamport.azurewebsites.net/video/videos.html

Key takeaway: how to describe a system as a state machine, check it with TLC, and state safety and liveness properties, using transaction commit and two-phase commit as the running examples.

2 Consensus · Paxos explained and specified

Paxos underpins modern distributed storage and consensus, but its original description was intentionally abstract. Lamport later clarified it into clean, practical principles.

Paper lamport.azurewebsites.net/pubs/paxos-simple.pdf

Tutorial & spec lamport.azurewebsites.net/tla/paxos-algorithm.html

Key takeaway: how Paxos is derived step by step from the safety requirement of consensus, and specified as three refinement levels that stay correct under lost, delayed, or reordered messages and crashed nodes.

3 TLA+ in action · Markus Kuppe's EWD998 workshop

Experience TLA+ in action by specifying distributed termination detection on a ring, due to Shmuel Safra. Each git commit introduces a new TLA+ concept, guided by Markus Kuppe (maintainer of the TLA+ tooling).

Repository github.com/tlaplus-workshops/ewd998

Key takeaway: how a short spec lets TLC find a false termination report that is hard to reproduce by testing, and how the spec grows step by step into a verified algorithm.

Suggested approach for engineering organizations

Prioritize reading invariants

The primary value comes from understanding and evaluating the conditions that must always hold, rather than memorizing specification syntax.

Translate specifications to PDFs

Render TLA+ specs as formatted PDFs for design reviews. A clean, mathematical spec is an unambiguous design document across engineering, architecture, and audit teams.

Inspect counterexample traces

Reviewing the error paths produced by the model checker builds shared intuition for rare failure modes.

Formal Systems

linkedin.com/in/formalsystems

We introduce the essentials in **3 days at your site in Europe/USA**, alongside tooling setup to manage **one million+ TLA+ specifications**, **TLAPS proofs**, and a **distributed setup** for both model checking and proof validation. Contact: ovidiu.marcu@formalstream.lu